

Executive Cars **(From Inspection to Auction, Everything smart)** **Smart AI Based Car Trading Platform**

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Submission Form for Final-Year

PROJECT REPORT



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NUMBER OF MEMBERS	3
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APPROVAL CERTIFICATE

This project, entitled as “Executive Cars” has been approved for the award of

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DECLARATION

I/We, hereby, declare that “No portion of the work referred to, in this project has been submitted in support of an application for another degree or qualification of this or any other university/institute or other institution of learning”. It is further declared that this undergraduate project, neither as a whole nor as a part thereof has been copied out from any sources, wherever references have been provided.

MEMBERS' SIGNATURES

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Executive Summary

Executive Cars is an innovative web platform designed to streamline the car selling and auctioning process while providing users with AI-driven insights into car pricing and comparisons. The project was developed to address the challenges faced by car sellers and buyers in evaluating, listing, and purchasing vehicles through a transparent and efficient system.

The platform consists of four core modules:

1. **Become A Seller:** Sellers schedule a car inspection at a designated branch. After inspection, the car is listed on the platform free of charge.
2. **Car Price Predictor:** An AI-powered tool predicts the estimated price of a car based on brand, model, mileage, and engine specifications using the Pak Wheels dataset (up to 2025).
3. **Auction Platform:** A subscription-based module where registered members gain access to curated auctions, real-time bidding, and detailed inspection reports.
4. **Already A Member:** Users with an active subscription can bid on cars.
5. **Become A Member:** New users sign up and complete payment via Stripe before accessing auctions.
6. **Used Cars:** Unsold auction cars can be listed as regular used cars at the seller's demand.

Additionally, the AI Car Assistant enables users to ask questions and compare vehicles. Executive Cars successfully integrates AI, real-time bidding, and secure payment solutions, making car trading more accessible and efficient.

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Chapter 1

Introduction

The automotive industry has long faced challenges such as inefficiencies, limited transparency, and difficulties in establishing fair market values. Conventional methods of buying and selling cars whether through dealerships, online platforms, or physical auctions often lead to trust concerns, inaccurate pricing, and restricted access to reliable vehicle information.

To overcome these issues, we propose Executive Car, an AI-powered, dual-tier online car auction platform designed to enhance transparency, efficiency, and accessibility within the automotive marketplace.

This chapter presents an overview of the Executive Car platform, outlining the key problems it addresses and the features that distinguish it from existing market solutions. It examines how the platform integrates AI-driven analytics, professional vehicle inspections, and a well-structured auction mechanism to deliver a streamlined experience for both buyers and sellers. Furthermore, it discusses the technologies employed in its development, defines the business scope, and details the allocation of project responsibilities.

By the conclusion of this chapter, readers will gain a clear understanding of the project's objectives, its technical framework, and the strategy for its implementation.

1.1. Project Introduction

1.1.1 Real-World Application

This project is developed for a real-world car showroom named **Executive Cars**. Currently, the showroom relies on traditional and manual methods for buying and selling vehicles, which often result in challenges such as limited transparency, inaccurate price estimation, and restricted customer reach.

To address these issues, the proposed system introduces a comprehensive digital solution that includes an AI-based car price prediction model, an online auction platform, and a system for verified vehicle listings with inspection reports. By integrating these features, the platform aims to enhance operational efficiency, ensure fair and data-driven pricing, and build greater trust between buyers and sellers.

Furthermore, this solution is expected to expand the showroom's market presence, improve customer engagement, and support overall business growth by transforming conventional processes into a modern, technology-driven ecosystem.

1.1.2 Background

In the automotive trading sector, the process of buying and selling vehicles is frequently hindered by inefficiencies and a lack of transparency, leading to dissatisfaction among both buyers and sellers. To overcome these challenges, we propose the development of an innovative two-tier online car trading platform. This solution is designed to transform the car trading experience by ensuring greater transparency, improved convenience, and enhanced efficiency for all stakeholders.

1.1.3 Problem Statement

The process of buying and selling cars continues to suffer from inefficiencies and a lack of transparency. Sellers often face difficulties in determining fair market prices, while buyers struggle to find trustworthy listings that include comprehensive inspection reports and transparent bidding processes. Traditional dealerships and auction systems frequently raise trust concerns, and online marketplaces carry additional risks due to inadequate verification mechanisms.

Furthermore, there is a lack of widely accessible AI-powered tools capable of accurately estimating vehicle prices using historical data. As a result, many sellers remain uncertain about the true value of their cars, leading to pricing inaccuracies and delays in transactions. The absence of a unified platform that integrates AI-based pricing, real-time auctions, and verified inspection reports highlights a significant gap in the market.

Executive Cars is designed to address these challenges by offering a seamless, transparent, and data-driven platform that enhances the experience for both buyers and sellers.

1.2. Existing Examples / Solutions

1.2.1 Examples

In Pakistan, while dedicated online car auction platforms may not be as prevalent as in some other countries, there are still several websites and platforms that facilitate the buying and selling of vehicles through auctions or listings.

1. **PakWheels:** PakWheels is one of the largest online automotive marketplaces in Pakistan. While it primarily functions as a classifieds platform for buying and selling vehicles, it also features auctions for certain listings.
2. **OLX Pakistan:** OLX is a popular online marketplace in Pakistan where users can buy and sell various goods, including vehicles. While it's not exclusively an auction platform, users can sometimes find vehicles listed for auction.

1.2.2 Solution

Executive Cars presents a digital car auction platform designed to improve transparency, efficiency, and accessibility within the automotive trading market. Its structured framework ensures a smooth and reliable experience for both buyers and sellers.

- **Ensuring Accurate Inspection:** Each vehicle undergoes a professional inspection, providing buyers with detailed and accurate reports along with clear pricing information, thereby building trust.
- **Streamlining Car Trade:** By digitizing the inspection, bidding, and transaction processes, Executive Cars significantly reduces the time and effort involved in car trading.
- **Vehicle Verification via Excise Records:** To further enhance transparency and trust, the system incorporates a simulated integration with official excise records using representative data to illustrate the vehicle verification process.

- **Securing transactions:** Registered paid members gain exclusive access to curated car auctions, ensuring availability of high-quality listings supported by verified reports.
- **Enhancing Market Access:** Sellers benefit from increased market exposure, while unsold vehicles can be redirected to a used car marketplace to improve their chances of sale.
- **Supporting Smart Decisions:** An integrated AI-based price prediction tool assists sellers in setting competitive prices, while an AI assistant provides real-time insights on vehicle comparisons, market trends, and maintenance guidance.
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1.2.3 Additional (optional) Feature:

- **Car Mechanic & Maintenance Support:** An additional feature can be integrated to provide users with car maintenance guidance and mechanic support. This module will offer basic troubleshooting, maintenance tips, and service reminders based on the vehicle's condition. Users can also connect with professional mechanics or get recommendations for servicing, ensuring better vehicle performance and long-term reliability.

Table 1.1: Existing Solutions vs Executive Cars

Feature	Pak Wheels	OLX Pakistan	Executive Cars
Car Listings	Comprehensive platform for listing vehicles	Free classified ads for vehicles	Comprehensive platform for listing vehicles
Featured Listings	Yes	Yes	Yes
Auction Platform	Occasionally hosts auction events	No	Yes
Vehicle Inspections	Offers professional inspection services	N/A	Yes
Community Forums	Hosts active community forums	N/A	No
News and Reviews	Provides news articles and reviews	N/A	No
Search and Filters	Search and filter options	Robust search and filter options	Yes
Direct Messaging	Facilitates communication between buyers/sellers	Provides messaging platform	No
Seller Verification	N/A	Offers seller verification features	Yes
Mobile App	Yes	Yes	No

1.3. Business Scope

The business scope of the proposed system covers multiple aspects of vehicle trading in Pakistan, focusing on improving the overall buying and selling experience. The key components of the business scope are outlined below:

- **Professional Inspection Services:** The platform will provide professional vehicle inspection services. Sellers can utilize these services to present accurate and reliable information about their vehicles, promoting transparency and building buyer confidence.
- **Seller Benefits:** Sellers can list their vehicles without any advertising costs, reducing their overall expenses. The platform handles the presentation of vehicle details, ensuring better visibility without extra effort. This allows sellers to focus on selling quickly and efficiently.
- **Buyer Benefits:** Buyers receive fully verified vehicles with detailed inspection reports and accurate condition information. This reduces the risk of fraud and builds trust in the purchase process. As a result, buyers can make confident and informed decisions.
- **Customer Engagement & Network Building:** The platform focuses on building long-term relationships with customers by keeping them connected through notifications, emails, and mobile updates. Users stay informed about new listings, offers, and relevant updates based on their interests. This continuous engagement helps retain customers and strengthens loyalty over time.
- **Premium Membership Model:** A premium membership option will be introduced for buyers, granting exclusive access to auction listings and advanced features. This model will serve as a revenue stream while delivering added value to users.
- **Maintenance & Growth Support:** The platform offers ongoing maintenance services to ensure smooth performance and a reliable user experience. Regular updates, system improvements, and customer support help maintain trust and satisfaction. These efforts not only support business growth but also strengthen and expand the customer network over time.

- **Comprehensive Vehicle Listings:** The platform will support detailed listings for a wide range of vehicles, including cars, motorcycles, and commercial vehicles. Sellers will be able to include specifications, condition details, pricing, and contact information.
- **Value-Added Services:** The platform may extend its offerings to include services such as vehicle financing, insurance solutions, and access to vehicle history reports, further strengthening its competitive advantage.
- **Future Expansion:** The system is currently being developed for local deployment within Islamabad. However, it is designed with scalability in mind, with future plans to expand its operation across Pakistan enabling wider access to a transparent and efficient car trading platform.

1.4. Useful Tools and Technologies

1.4.1. Languages:

1. JavaScript:

- JavaScript is utilized as the core programming language for both frontend and backend development within the project. It facilitates the creation of dynamic and interactive user interfaces on the frontend while also supporting server-side logic on the backend.

1.4.2. Frameworks:

Frontend Development:

1. React with Tailwind CSS:

- React is a widely used JavaScript library for developing user interfaces, particularly for single-page applications. It allows developers to build reusable components and efficiently manage updates through the virtual DOM. Tailwind CSS, a utility-first CSS framework, streamlines the styling process by offering predefined classes for building responsive and

modern designs. Together, React and Tailwind CSS enhance development efficiency, code maintainability, and overall user experience.

1.4.3. Backend Development:

1. Express.js with Node.js:

- Express.js, a lightweight web application framework for Node.js, is used to develop the backend of the system. Node.js enables the execution of JavaScript on the server side, while Express.js simplifies the creation of scalable and robust web applications through its extensive features and middleware support.

1.4.4. System Requirements:

1. Operating Systems:

- The application is designed to be compatible with major operating systems, including Windows, macOS, and Linux, ensuring accessibility for a diverse range of users.

2. Hardware Requirements:

- A minimum of 4GB RAM is recommended to ensure smooth system performance.
- At least 32GB of storage is required to support the application and its associated data.

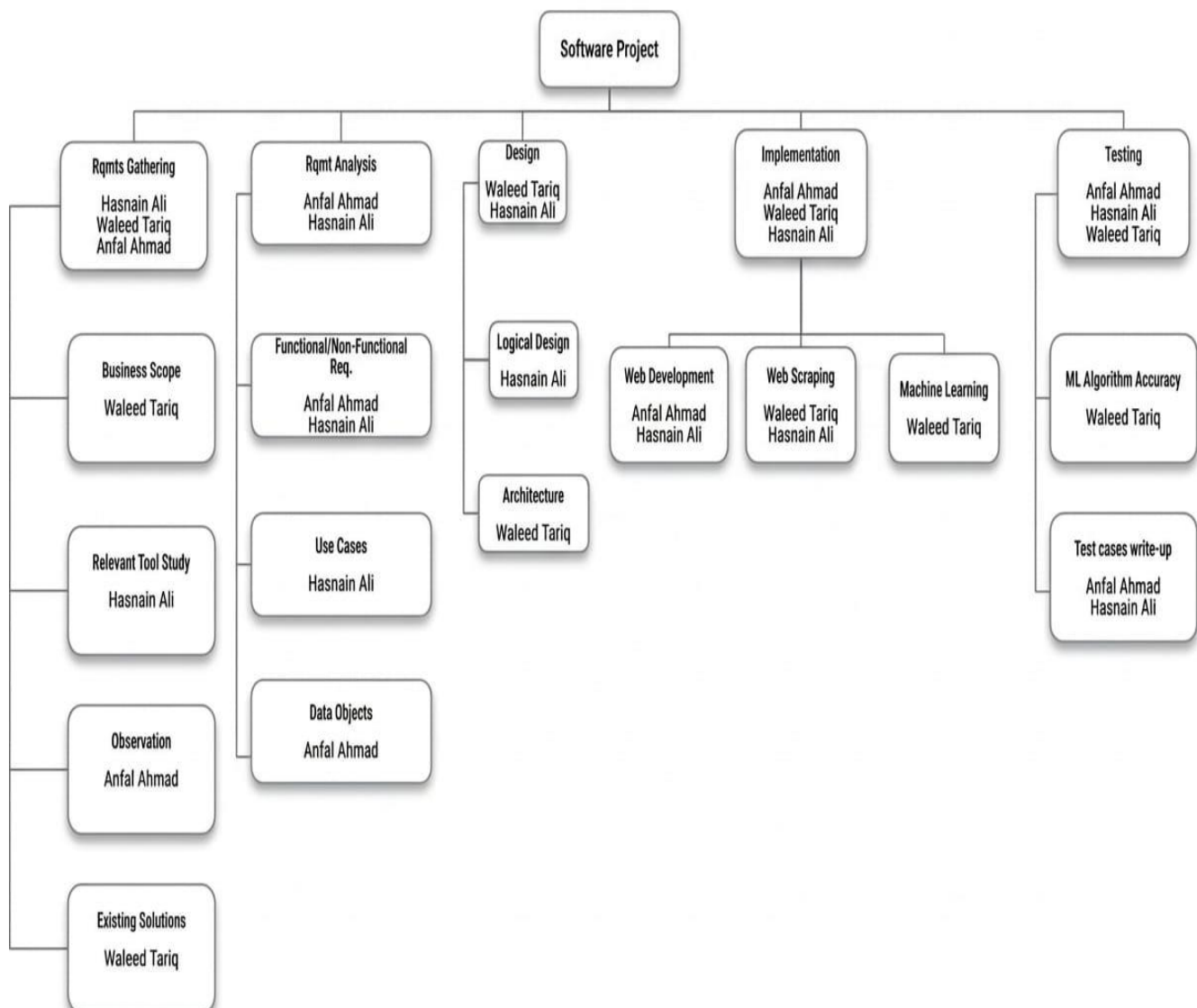
3. Internet Connection:

- A stable and reliable internet connection is necessary for accessing the platform and utilizing its features effectively.

1.5. Project Work Break Down

The project is divided into multiple components, each assigned to team members based on their respective areas of expertise. A significant portion of the timeline has been dedicated to web development, as building and integrating both the frontend and backend requires considerable effort, particularly given the expanded scope of the project.

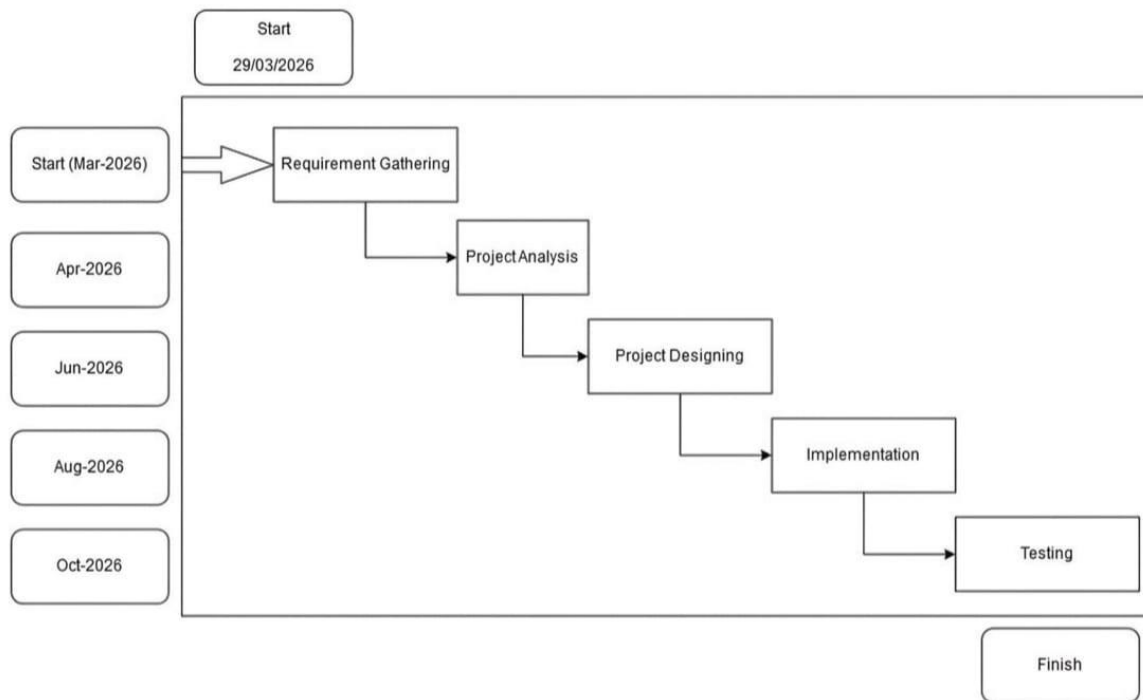
Figure 1.1: Project Work breakdown Structure



1.6. Project Time Line

The project time line has been assigned thoughtfully as the team acknowledges the various modules full potential towards their end.

Figure 1.2: Executive Cars Project Timeline



1.7. Conclusion

As this chapter concludes, the business aspect of the project and the technical aspect have been understood including the tech stack that will be used for developing the project. The project has been divided into 2 sections. The web development, both front-end and back-end, has been assigned to Muhammad Hasnain and Anfal Ahmad, whereas the Machine Learning modules has been assigned to Waleed Tariq.

Figure 1.3 Project Workflow Diagram

